

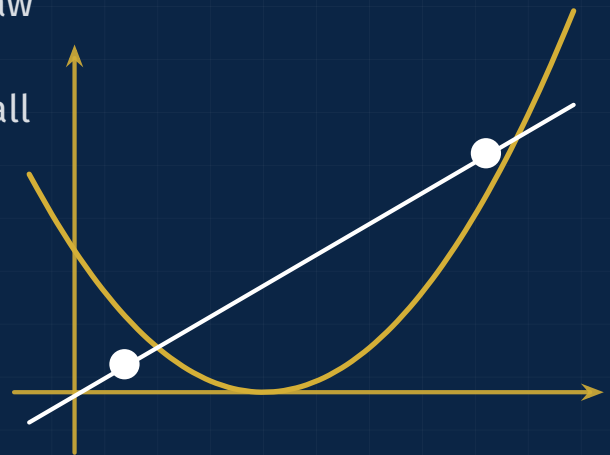


DR ESLAM AHMED
eliteigcse.com

CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 8

Graphing Inequalities

Each inequality is a line and a side. Draw the lines (solid or dashed), test a point to pick the side, and the region where all sides overlap is R.



3

VARIANTS

19

PAST-PAPER PARTS

22

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Draw & shade a region

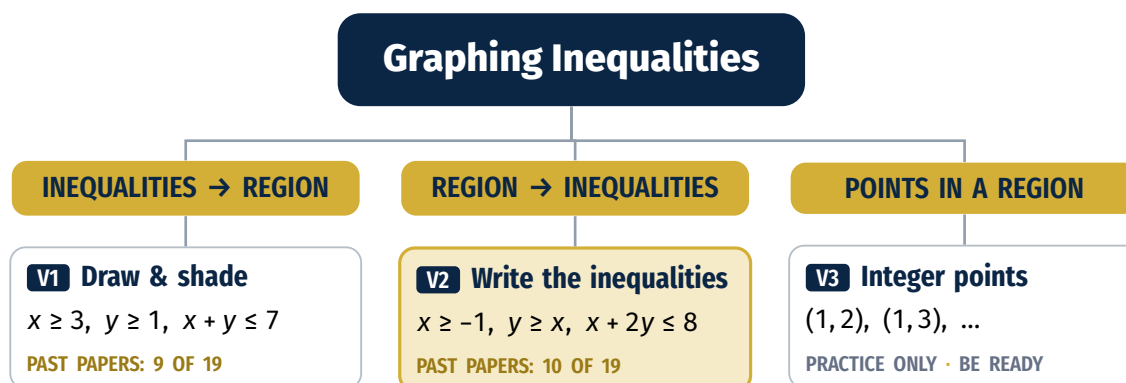
V2 Write the inequalities

V3 Integer points in a region

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Graphing inequalities at a glance



Your notes
**THE KEY HABIT****TEST A POINT**

Put a point inside R into each inequality — all must be true.

★ How often does each variant come up?

15 past-paper questions (Jan 2020 → Nov 2025) hold 19 parts on regions; 22 practice parts — grouped by type.

	PAST PAPERS (OUT OF 19)	PRACTICE PARTS
V1 Draw, shade a region	9	14
V2 Write the inequalities	10	6
V3 Integer points	0	2

WHAT THIS TELLS YOU On almost every paper, early (Q2–Q13, 3–4 marks): either shade R from three inequalities, or read three inequalities from a shaded R. Always three lines, always a triangle.

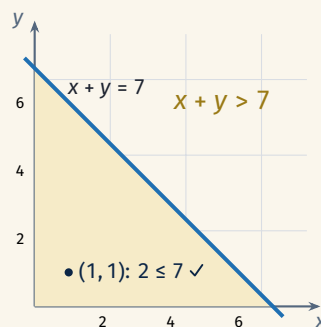
BEFORE YOU START

The big idea

A line splits the grid into two sides.

$x + y = 7$ is the line. $x + y \leq 7$ is the line and everything below-left of it.

Three inequalities $\rightarrow$ three lines $\rightarrow$ the **overlap** of the three sides is the region R.



Line types and sides

$\leq$ or $\geq$

solid line

The line is included.

$<$ or $>$

dashed line

The line is not included.

$y \geq \dots$

above

When y is alone.

$x \leq \dots$

left

Vertical line, left side.

Draw a line fast

$x = 3$

vertical

Through 3 on the x -axis.

$y = 1$

horizontal

Through 1 on the y -axis.

$x + y = 7$

$(0, 7), (7, 0)$

Cover-up: $x = 0$, then $y = 0$.



CALCULATOR FIRST, THEN CHECK

Test point: pick a point clearly inside (not on a line), substitute into all three — all true means you shaded the right region.



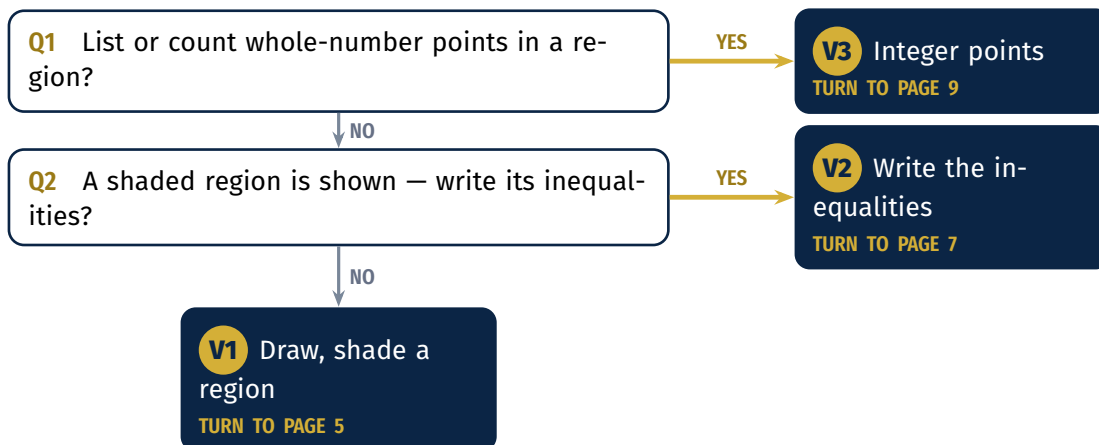
Your notes

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check

V1

VARIANT 1

Draw & Shade a Region

Draw each line, decide each side with a test point, shade where all three overlap, label R.

SEEN IN PAST PAPERS

9 / 19

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2021 P2H Q4 • Jan 2022 P2HR Q7(b) • Jun 2022 P2H Q2 • Jun 2024 P2H Q2 • Nov 2024 P2H Q7

? What is it?

In plain words

Draw the three **boundary lines** and label them. For each inequality decide the **correct side**.

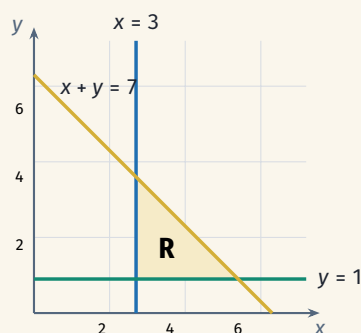
R is the part that is on the right side of **all three** lines — usually a triangle. Shade it and write **R** inside.

IT LOOKS LIKE

$x \geq 3$

$y \geq 1$

$x + y \leq 7$



How do I do it?

1 Lines

Draw and label each boundary (solid/dashed).

2 Sides

$x \geq 3$: right; $y \geq 1$: above; $x + y \leq 7$: test (0,0).

3 Overlap

The region on the correct side of all three.

4 Label

Shade it; write R.

LINE, SIDE, OVERLAP Then label R.

V1 Draw & Shade a Region

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“On the grid, draw and label the straight line ...”

→ **Vertical, horizontal, or cover-up**

“Show, by shading ... the region that satisfies all three”

→ **Overlap of the three sides**

“Label the region R”

→ **Write R inside — a mark of its own****WORKED EXAMPLE****PAST PAPER**

November 2024 · Paper 2H · Q7 · 4 marks

(a) Draw and label $x = 3$, $y = 1$, $x + y = 7$. (b) Shade the region that satisfies $x \geq 3$, $y \geq 1$, $x + y \leq 7$. Label it R.

STEP 1 $x = 3$, $y = 1$; $x + y = 7$ through $(0, 7)$, $(7, 0)$ ◀ (a) the lines

STEP 2 $x \geq 3$: right; $y \geq 1$: above ◀ sides

STEP 3 $(0, 0)$: $0 \leq 7$ true $\Rightarrow$ below $x + y = 7$ ◀ test point

STEP 4 triangle $(3, 1)$, $(6, 1)$, $(3, 4)$ ◀ (b) overlap; label R

ANSWER triangle with corners $(3, 1)$, $(6, 1)$, $(3, 4)$ ✓ $(4, 2)$: $4 \geq 3$, $2 \geq 1$, $6 \leq 7$ ✓

**WATCH OUT**

✗ $x \geq 3$ shaded below

✓ to the right

$x \geq$: larger x values, so to the right of the vertical line.

**EXAM TIP**

Write the corner coordinates on the diagram — they show the examiner your region is exact.

**Check yourself**

calculator first · answers on page 11

1.1

Jun 2022 P2H

Region $y \geq 1$, $x \geq 2$, $x + y \leq 7$. Give the corners of R.

Answer:

1.2

Jun 2024 P2H

Region $y \geq 2$, $x \leq 6$, $y \leq x + 1$. Give the corners of R.

Answer:

1.3

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Region $x \geq 1$, $y \geq 2$, $x + y \leq 6$. Give the corners of R.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

10 / 19

PRACTICE QUESTIONS: 6

Write the Inequalities for a Region

Name each boundary line, then use a point inside R to choose $\leq$ or $\geq$.

Came up in: Jan 2020 P1H Q10 • Jan 2020 P2HR Q11(b) • Jan 2021 P2HR Q7 • Nov 2021 P2H Q13 • Jan 2022 P2H Q13 • May 2023 P1H Q7 • Jun 2023 P2HR Q8(b) • Nov 2023 P1H Q8(b) • Jun 2025 P2H Q9(b) • Nov 2025 P2H Q10(b)

 Your notes

? What is it?

In plain words

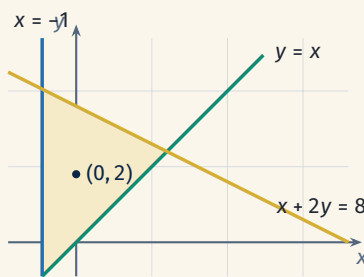
Work backwards. First the **equation of each line**: vertical $x = \dots$, horizontal $y = \dots$, sloped $y = mx + c$ (read m and c).

Then pick a point **inside R** and use it to choose the sign for each line.
Solid lines: $\leq, \geq$.

IT LOOKS LIKE

three lines, one given

R shaded



How do I do it?

1 Lines

Equation of each boundary.

2 Point

Choose a point clearly inside R.

3 Signs

Put it in each: true with $\geq$ or $\leq$?

4 Write

Three inequalities (solid: $\leq, \geq$).

TEST A POINT

The most common question here.

V2 Write the Inequalities for a Region

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Write down the three inequalities that define the shaded region” → **Equations, then test a point**

A sloped line with no label → **Read m and c from the grid**

Line through the origin at 45° → $y = x$

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 2H • Q13 • 3 marks

The shaded region is bounded by $x = -1$, the line through $(0, 0)$ and $(2, 2)$, and $x + 2y = 8$. The point $(0, 2)$ is inside it. Write down three inequalities that define the region.

STEP 1 $x = -1, y = x, x + 2y = 8$ ◀ the three lines

STEP 2 $(0, 2) : 0 \text{ vs } -1 \Rightarrow x \geq -1$ ◀ test point

STEP 3 $(0, 2) : 2 \text{ vs } 0 \Rightarrow y \geq x$ ◀ test point

STEP 4 $(0, 2) : 4 \text{ vs } 8 \Rightarrow x + 2y \leq 8$ ◀ test point

ANSWER $x \geq -1, y \geq x, x + 2y \leq 8$ ✓ all three true at $(0, 2)$ ✓

WATCH OUT

✗ $y \leq x$ for the region above $y = x$

✓ $y \geq x$

Above a line means bigger y .

EXAM TIP

The test point settles every sign in 10 seconds — never guess.

Check yourself

calculator first • answers on page 11

2.1 Jun 2025 P2H

R is the triangle with corners $(2, 3), (6, 3), (2, 7)$. Write its three inequalities.

Answer:

2.2 Nov 2023 P1H

R is the triangle with corners $(2, 2), (7, 2), (7, 7)$. Write its three inequalities.

Answer:

2.3 Nov 2025 P2H

R is the triangle with corners $(-3, 4), (4, 4), (-3, -3)$. Write its three inequalities.

Answer:



VARIANT 3

Integer Points in a Region

List the whole-number points column by column and test each one.

SEEN IN PAST PAPERS

0 / 19

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 2



Your notes

What is it?

In plain words

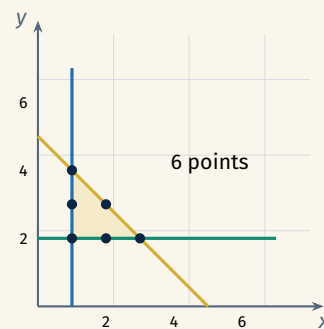
Integer points have **whole-number** coordinates. Go **column by column** ($x = 1$, then $x = 2$, ...) and list every y that works.

Points **on** a solid line count; on a dashed line they do **not**.

IT LOOKS LIKE

list the integer points

greatest $x + y$



How do I do it?

1 Range of x

Smallest to largest x in R .

2 Each column

Which whole y values fit?

3 Edges

Solid: include. Dashed: exclude.

4 List

Write every pair; count if asked.

COLUMN BY COLUMN No point gets missed.

V3 Integer Points in a Region

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“List the points with integer coordinates in R ”

→ **Column by column**

Strict inequalities ($<$, $>$)

→ **Points on those lines don't count**

“the greatest value of $x + y$ ”

→ **Test the corner points****WORKED EXAMPLE DR ESAM**

Integer points • 3 marks

R is defined by $x \geq 1$, $y \geq 2$, $x + y \leq 5$. List all the points with integer coordinates in R .

STEP 1 $x = 1 : y = 2, 3, 4$ ◀ $y \leq 4$

STEP 2 $x = 2 : y = 2, 3$ ◀ $y \leq 3$

STEP 3 $x = 3 : y = 2$ ◀ $y \leq 2$

STEP 4 $x = 4 : y \leq 1$ — none ◀ stop

ANSWER $(1, 2), (1, 3), (1, 4), (2, 2), (2, 3), (3, 2)$

✓ 6 points ✓

WATCH OUT

✗ leaving out $(1, 4)$ on the line

✓ include it

$x + y \leq 5$ includes the line: $1 + 4 = 5$.

EXAM TIP

Count the points in each column first, then write the list.

Check yourself

calculator first • answers on page 11

3.1

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$R: x > 1, y > 1, x + y < 6$.
List the integer points in R .

Answer:

3.2

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Is $(2, 3)$ in the region
 $y \geq x, x + y \leq 6, x \geq 0$?

Answer:

3.3

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$R: x \geq 3, y \geq 1, x + y \leq 7$.
Find the greatest value
of $x + y$ at an integer
point.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 (2, 1), (6, 1), (2, 5)

1.2 (1, 2), (6, 2), (6, 7)

1.3 (1, 2), (4, 2), (1, 5)

2.1 $x \geq 2$, $y \geq 3$, $x + y \leq 9$

2.2 $y \geq 2$, $x \leq 7$, $y \leq x$

2.3 $x \geq -3$, $y \leq 4$, $y \geq x$

3.1 (2, 2), (2, 3), (3, 2)

3.2 Yes

3.3 7

THE WHOLE TOPIC ON ONE PAGE

Inequalities on graphs cheat sheet

**V1 Draw & shade 9/19**

$$x \geq 3, y \geq 1, x + y \leq 7$$

Line, side, overlap, label R.

V2 Write them 10/19

$$x \geq -1, y \geq x$$

Equations, then test a point.

V3 Integer points

(1, 2), (1, 3), ...

Column by column; solid edges count.

**TOP 3 LOST MARKS**

1. wrong side of a line ×
2. R not labelled ×
3. < written for a solid line ×

BOTTOM LINE Solid $\leq, \geq$, dashed $< >$ • test a point • overlap of all three • label R.